# 1. Hospital Overview

Admissions, occupancy, stay length, readmission. Filters on top apply to every page.

**CareLens**

Hospital operations & patient analytics · Jul 2019 – Apr 2024

Overview | Patient flow | Departments | Resources

From: Jul 2019 | To: Apr 2024 | Department: All | Admission type: All | Reset

53,178 admissions in all departments from Jul 2019 to Apr 2024. Beds were 86.4% full on average and a typical stay lasted 15.5 days.

| Total admissions | Occupancy rate | Average length of stay | 30-day readmission | Bed utilization | Discharges |
|---|---|---|---|---|---|
| 53,178 | 86.4% | 15.5 d | 0.1% | 91.8% | 53,115 |
| 917 per month | target 85% | days per stay | see note below | of staffed beds | patients sent home |

**Admissions per month**

How busy the hospital was




**Occupancy per month**

Share of licensed beds in use




**Admissions by department**

Selected months

| | |
|---|---|
| Endocrinology | 17,744 |
| Rheumatology | 8,933 |
| Cardiology | 8,875 |
| Oncology | 8,837 |
| Pulmonology | 8,789 |

**How patients arrive**

Elective, emergency, urgent (ignores the admission-type filter)

| | |
|---|---|
| Elective | 33.5% |
| Emergency | 33.0% |
| Urgent | 33.5% |

**Average stay by department**

Days

| | |
|---|---|
| Cardiology | 15.4 |
| Endocrinology | 15.4 |
| Oncology | 15.5 |
| Pulmonology | 15.7 |
| Rheumatology | 15.5 |

Note: the dataset has no patient ID, so a “readmission” is a new admission within 30 days for someone with the same name, gender and blood type. That is why the rate is so low.

Admissions come from the Kaggle “Healthcare Dataset”. Beds, staff, equipment and patient feedback are simulated supporting data (see docs/Dataset_Sources.md). When an admission type is selected, occupancy shows the share of beds used by that type.

# 2. Patient Flow

Busiest days, peak load, discharges and how long stays last.

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The busiest admission day is Tue. At its peak, 552 patients were in beds on one day (102.2% of licensed beds).

### Admissions vs discharges

Per month



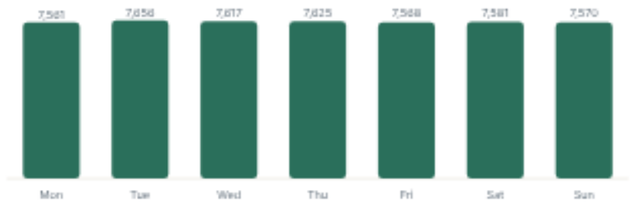

### Peak patient load

Highest number of patients in beds on any day of the month




### Admissions by weackday

Selected months



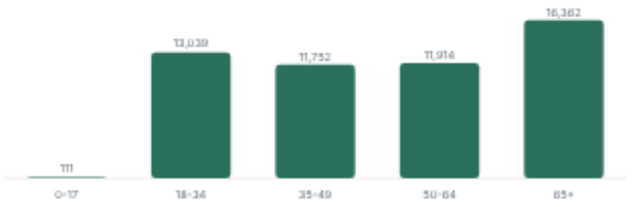

### Length of stay

Patients by stay length (days)

| | |
|---|---|
| 1–7 | 12,450 |
| 8–14 | 12,454 |
| 15–21 | 12,393 |
| 22–30 | 15,881 |

### Age groups

Patients admitted




### Test results

Share of patients

| | |
|---|---|
| Normal | 33.4% |
| Abnormal | 33.5% |
| Inconclusive | 33.1% |

Admissions come from the Kaggle “Healthcare Dataset”. Beds, staff, equipment and patient feedback are simulated supporting data (see docs/Dataset_Sources.md). When an admission type is selected, occupancy shows the share of beds used by that type.

# 3. Department Analytics

Volume, readmission and the efficiency score for each department.

**CareLens**

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Overview | Patient flow | **Departments** | Resources

From: Jul 2019 | To: Apr 2024 | Department: All | Admission type: All | Reset

Oncology scores highest on efficiency (87.3 out of 100) for Jul 2019 to Apr 2024. Click a row to focus the other pages on one department.

| Department | Admissions | Avg stay | Readmit | Occupancy | Bed use | Satisfaction | Efficiency |
|---|---|---|---|---|---|---|---|
| Cardiology | 8,875 | 15.4 | 0.1% | 86.0% | 91.7% | 4.2 | **83.7** |
| Endocrinology | 17,744 | 15.4 | 0.1% | 86.1% | 91.4% | 4.2 | **86.3** |
| Oncology | 8,837 | 15.5 | 0.1% | 86.3% | 91.6% | 4.2 | **87.3** |
| Pulmonology | 8,789 | 15.7 | 0.1% | 86.8% | 92.3% | 4.2 | **81.9** |
| Rheumatology | 8,933 | 15.5 | 0.1% | 86.9% | 92.2% | 4.2 | **82.7** |

**Efficiency score**

0–100 · 40% stay length vs target, 30% bed use near 85%, 30% satisfaction

| | |
|---|---|
| Cardiology | 83.7 |
| Endocrinology | 86.3 |
| Oncology | 87.3 |
| Pulmonology | 81.9 |
| Rheumatology | 82.7 |

**Patient volume**

Admissions in the selected months

| | |
|---|---|
| Cardiology | 8,875 |
| Endocrinology | 17,744 |
| Oncology | 8,837 |
| Pulmonology | 8,789 |
| Rheumatology | 8,933 |

**Readmission by department**

30-day readmission rate

| | |
|---|---|
| Cardiology | 0.1% |
| Endocrinology | 0.1% |
| Oncology | 0.1% |
| Pulmonology | 0.1% |
| Rheumatology | 0.1% |

**Treatment capacity**

Licensed beds per department

| | |
|---|---|
| Cardiology | 90 |
| Endocrinology | 180 |
| Oncology | 90 |
| Pulmonology | 90 |
| Rheumatology | 90 |

Admissions come from the Kaggle "Healthcare Dataset". Beds, staff, equipment and patient feedback are simulated supporting data (see docs/Dataset_Sources.md). When an admission type is selected, occupancy shows the share of beds used by that type.

# 4. Resource Utilization

Beds, staff and equipment use, and where capacity is tight.

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Overview | Patient flow | Departments | **Resources**

From: Jul 2019 | To: Apr 2024 | Department: All | Admission type: All | Reset

Pulmonology runs the tightest, using 92.3% of its staffed beds. Beds, staff and equipment figures on this page are simulated.

| Bed utilization | Nurses per 10 patients | Equipment utilization | Avg wait to be seen |
|---|---|---|---|
| **91.8%** | **5.2** | **73.1%** | **43 min** |
| of staffed beds | nurses on the roster | hours used / available | from feedback data |

### Bed utilization by department

Orange mark = 85% target

| | |
|---|---|
| Pulmonology | 92.3% |
| Rheumatology | 92.2% |
| Cardiology | 91.7% |
| Oncology | 91.6% |
| Endocrinology | 91.4% |

### Equipment utilization

Hours used / hours available

| | |
|---|---|
| Pulmonology | 73.6% |
| Rheumatology | 73.4% |
| Cardiology | 72.5% |
| Oncology | 73.5% |
| Endocrinology | 72.4% |

### Staffed vs licensed beds

Beds that actually have staff

| | |
|---|---|
| Pulmonology | 94.0% |
| Rheumatology | 94.2% |
| Cardiology | 93.8% |
| Oncology | 94.2% |
| Endocrinology | 94.2% |

### Months above 90% occupancy

Capacity planning: where to add beds first

| | |
|---|---|
| Cardiology | 18 mo |
| Endocrinology | 13 mo |
| Oncology | 18 mo |
| Pulmonology | 21 mo |
| Rheumatology | 20 mo |

### Patient satisfaction

Average score out of 5

| | |
|---|---|
| Pulmonology | 4.2 |
| Rheumatology | 4.2 |
| Cardiology | 4.2 |
| Oncology | 4.2 |
| Endocrinology | 4.2 |

### Occupancy by month

Selected departments combined




Admissions come from the Kaggle "Healthcare Dataset". Beds, staff, equipment and patient feedback are simulated supporting data (see docs/Dataset_Sources.md). When an admission type is selected, occupancy shows the share of beds used by that type.